Kartik Loya

Robotics & Controls | Model Predictive Control | Physical Robot Testing

Clemson, SC, USA | Kloya@clemson.edu | linkedin.com/in/kartikloya | github.com/kloya03

Summary

Robotics and controls researcher with hands-on experience building, integrating, and field-testing physical robots. Developed and deployed real-time model predictive control (MPC) and learned controllers for trajectory tracking, achieving 1.8–6.3 cm cross-track RMSE on an underwater robot. Experience spans vehicle dynamics, localization and state estimation, sensor and actuator integration, and sim-to-real learning. Mechanical Engineering Ph.D. expected in 2026; programming in C++, Python, and MATLAB.

Robotics Research Experience

Graduate Research Assistant, Virtual Prototyping of Ground Vehicle Systems Center (VIPR-GS) – Clemson, SC, USA — Sept 2024 – present

Advisor: Prof. Phanindra Tallapragada

`Data-driven Koopman Control-Oriented Learning Framework for Improved Autonomous Offroad Mobility`

- Integrated learned vehicle dynamics with constrained MPC for aggressive off-road trajectory tracking under steering and torque limits, achieving 85 ms per optimization step.
- Currently integrating the framework on a 1/10th-scale autonomous off-road vehicle with a depth camera, wheel encoders, IMU, and GPS for onboard perception, state estimation, and closed-loop control.
- Built fast, terrain-specific vehicle-dynamics predictors from large-scale simulation data using physics-informed Koopman learning for control on deformable terrain.
- Evaluated dynamics models across unseen soil conditions, identifying performance degradation that motivates terrain-adaptive modeling and control.

Graduate Research Assistant, Applied Dynamical Systems Lab – Clemson, SC, USA — Aug 2021 – May 2024

Advisor: Prof. Phanindra Tallapragada

- Deployed real-time MPC and learned controllers on a physical underwater robot, achieving 1.8–6.3 cm cross-track RMSE and speeds up to 2 body lengths/s with low-level BLDC motor and servo control.
- Designed, built, and field-tested the robot in pool and lake environments; integrated internal-rotor propulsion, a servo-actuated tail, onboard compute, IMU, pressure sensors, GPS, and overhead vision-based marker tracking for localization and state estimation.
- Developed a physics-based nonlinear simulator capturing hydrodynamics, underactuation, and coupled body–tail dynamics for scalable data generation, system identification, and controller development.
- Built sim-to-real learning pipelines using Koopman learning and differentiable reinforcement learning (RL), combining large-scale simulation with limited real-world data to learn transferable dynamics models and control policies.
- Developed online model adaptation that selectively incorporates informative streaming sensor data, supporting learning across changing dynamical regimes while reducing redundant data and computation.
- Analyzed roll stability under periodic body motion and fluid interaction, identifying operating conditions that amplify roll oscillations.

Graduate Research Assistant, Virtual Prototyping of Ground Vehicle Systems Center (VIPR-GS) – Clemson, SC, USA Jan 2020 – Aug 2021

Advisor: Dr. Mohammad Naghnaeian

- Energy management of multi-scale vehicle fleets
- Designed energy-efficient planning and control strategies for UAV fleets operating in resource-constrained environments.
- Applied MPC to UAV energy management, accounting for disturbances, wind effects, and morphed UAV configurations.

Technical Skills

Programming & Robotics Software: C++, Python, MATLAB; ROS, ROS 2, OpenCV, PyTorch

Controls & Modeling: Model predictive control (MPC), trajectory tracking, system identification, vehicle dynamics, state estimation, online model adaptation, sim-to-real learning

Planning & Localization Coursework: Coursework experience with motion-planning and localization algorithms for off-road vehicle applications

Simulation & Optimization: Project Chrono, Isaac Sim, MuJoCo, CasADi, IPOPT, GPOPS-II, SolidWorks, Mathematica

Sensors & Localization: IMU, GPS, wheel encoders, pressure sensors, RGB/depth cameras, LiDAR, DepthAI OAK, Intel RealSense, vision-based marker tracking

Embedded Systems & Tools: Linux, Git, Docker, Raspberry Pi, Radxa, NVIDIA Jetson Nano, ESP32

Actuation & Communication: BLDC motors, servo motors, Tinymovr motor drivers, encoder feedback, CAN, UART, I2C, SPI, RF communication

Engineering Experience

Engineering Intern, Thermax Ltd. – Pune, India Oct 2017 – May 2018

- Developed a Python program for remote online performance analysis of Vapour Absorption chiller plants.
- Designed and manufactured a boiler panel cutting machine to replace flame cutting and avoid heat-affected zones during boiler repair.
- Surveyed and optimized the steel utilization in the plant.

Engineering Intern, Elite Techno Group – Pune, India June 2016

- Designed an ATV in SolidWorks, performed crash analysis, and manufactured and tested the vehicle on a road track.

Education

Clemson University, Ph.D. in Mechanical Engineering – Clemson, SC, USA 2021 - 2026 (expected)

- GPA: 4.00/4.00

Clemson University, M.S. in Mechanical Engineering – Clemson, SC, USA 2018 - 2021

- GPA: 3.87/4.00

Maharashtra Institute of Technology, Pune University, B.E. in Mechanical Engineering – Pune, India 2014 - 2018

Competitions

AI for Industry Challenge - Intrinsic × Open Robotics – Team ASIMOV May 2026 – Aug 2026

- Ranked 21st globally and advanced to Phase 1 (Top 30 of 160+ teams) in a worldwide robotics competition focused on autonomous manipulation of deformable cables and precision connector insertion.
- Developed the manipulation solution using Intrinsic Flowstate, integrating Vision-Language-Action (VLA) models, reinforcement learning, and OpenCV-based perception for task planning, visual understanding, and robot control.
- Designed and tested reusable robot skills for perception and manipulation within the Flowstate workflow, combining learning-based approaches with task-specific robotic behaviors.

iGRADS (Graduate Research and Discovery Symposium) Mar 2026 – Apr 2026

- Advanced to the final round of Clemson's university-wide graduate research competition, presenting doctoral research to a broad, nontechnical audience.

Honors and Awards

ME Excellence Award - Ph.D. – Clemson University Apr 2026

- Recognized by the Department of Mechanical Engineering for excellence as a Ph.D. graduate student.

Graduate Travel Grant – Clemson Graduate Student Government Jan 2024

- Awarded competitive graduate travel funding to attend the IEEE International Conference on Robotics and Automation (ICRA 2024).

Publications

Learning Koopman operators for path tracking by a fish-like robot
Kartik Loya, P. Chivkula, P. Tallapragada
(Submitted to IEEE Transactions on Robotics)

Differentiable Reinforcement Learning for Path Tracking by an Agile Fish-Like Robot Jan 2026
Kartik Loya, P. Chivkula, R. Varikuti, P. Tallapragada
(IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS))

Koopman Operator Framework for Modeling and Control of Off-Road Vehicle on Deformable Terrain Jan 2025
Kartik Loya, P. Tallapragada
(ASME Journal of Autonomous Vehicles and Systems (JAVS))

Agile 3-D Motion Primitives in an Underactuated Fish-like Robot Jan 2025
P. Chivkula, *Kartik Loya*, P. Tallapragada
(IEEE Robotics and Automation - Letters)

Parametric roll oscillations of a hydrodynamic Chaplygin sleigh Jan 2023
Kartik Loya, P. Tallapragada
(Nonlinear Dynamics 111 (22), 20699-20713)

Online learning of Koopman operator using streaming data from different dynamical regimes Jan 2024
Kartik Loya, P. Tallapragada
(IFAC-PapersOnLine 58.28, 90-95)

Koopman operator-based predictive control uses a data archive of observables Jan 2023
Kartik Loya, J. Buzhardt, P. Tallapragada
(ASME Letters in Dynamic Systems and Control 3 (3))

Stabilization of an Inverted Pendulum on a Nonholonomic System Jan 2022
Kartik Loya, P. Tallapragada
(IFAC-PapersOnLine 55 (37), 764-769)

Talks and Presentations

- **Modeling, Estimation and control Conference (MECC)** — 2022: "Stabilization of an Inverted Pendulum on a Nonholonomic System"
- **Modeling, Estimation and control Conference (MECC)** — 2023: "Koopman Operator Based Predictive Control With a Data Archive of Observables"
- **Modeling, Estimation and control Conference (MECC)** — 2024: "Online learning of Koopman operator using streaming data from different dynamical regimes"
- **SIAM Conference on Applications of Dynamical Systems (DS25)** — Talk: Model-based learning of Koopman operator for complex mechanical systems (May 2025).
- **ISTVS 2025 (55th Conference)** — Talk: Data-driven Koopman control-oriented learning framework for improved autonomous off-road mobility (Oct 9, 2025).
- **ICRA 2025** — Poster: "Batch learning/updates of Koopman operators from streaming off-road vehicle data with terrain dynamics" (Summer 2025).
- **ICRA 2025** — Late Breaking Results: "Learning PID Gains for Planar Tracking of Bio-Inspired Swimming Robots Using Simplified Models"
- **Clemson VIPR Annual Meeting 2025** — Poster Presentation: "Koopman-Based Modelling and Control of Off-Road Vehicles on Deformable Terrain"
- **Clemson Graduate Student Seminar (GSR)** — Research talk: 2023 "Parametric Roll Oscillation of a Hydrodynamic Chaplygin Sleigh"
- **Clemson Graduate Student Seminar (GSR)** — Research talk: 2025 "Data-driven Koopman Control-Oriented Learning Framework for Improved Autonomous Off-road Mobility."

Coursework

Graduate Coursework: Intermediate Dynamics | Nonlinear Dynamics & Chaos | Robot Multibody Dynamics | Advanced Nonlinear Dynamics | Advance Engineering Mathematics | Linear Analysis | Control Systems Engineering | Advanced Nonlinear control | Random Processes and Application | Applied Optimal Control | Networked Dynamical systems | Advanced Estimation | Control of Cyber-Physical Systems | Autonomy: Science & Systems | Reinforcement Learning | Advanced Machine Learning | Theory of Elasticity